VNIT Nagpur

Department of Electronics and Communication

2024-2025

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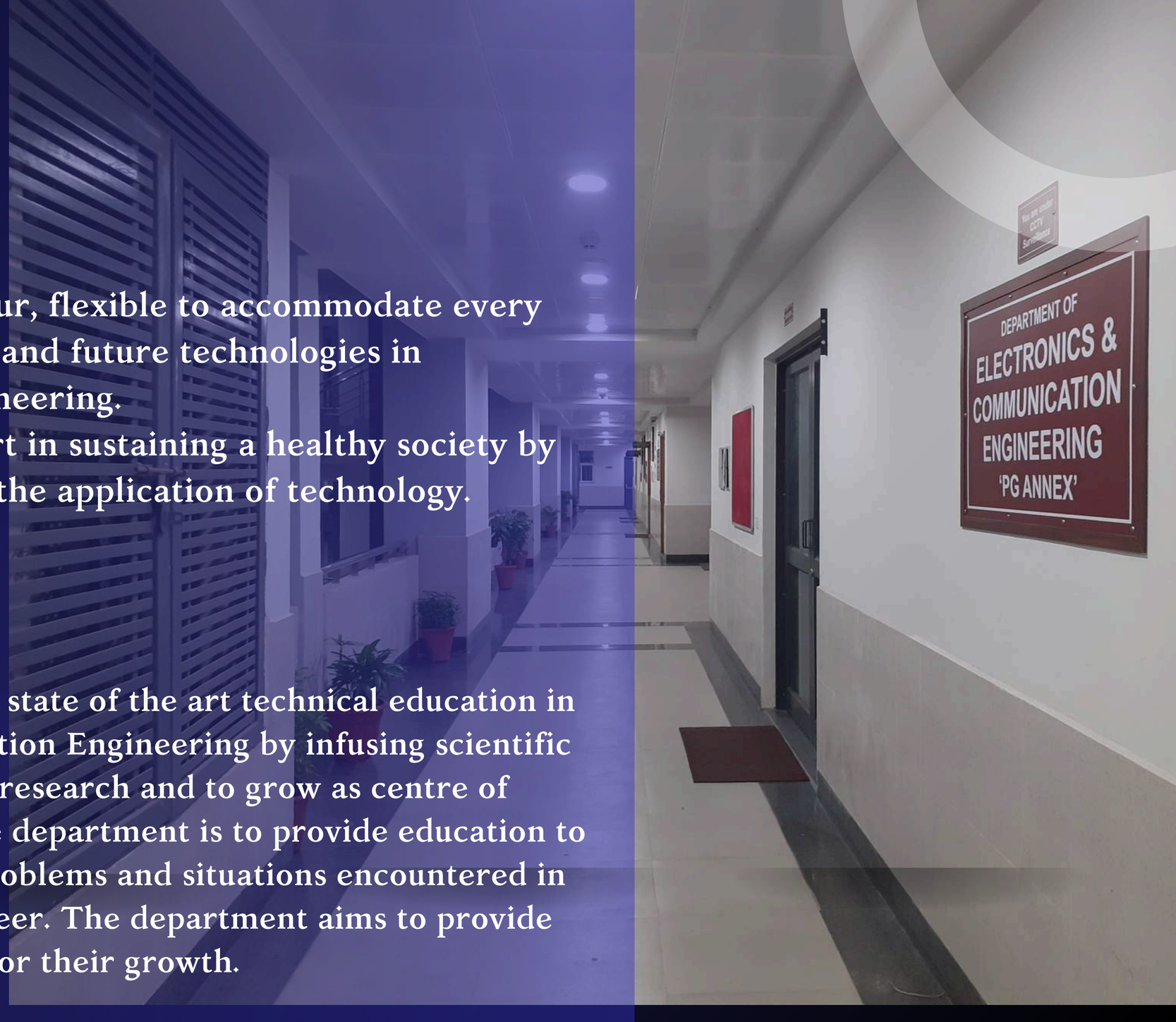
ABOUT US

MISSION

1.To be the epitome of academic rigour, flexible to accommodate every student and faculty for basic, current and future technologies in Electronics and Communication Engineering.
2.Strengthening and providing support in sustaining a healthy society by improving the quality of life through the application of technology.

VISION

The Department endeavors to facilitate state of the art technical education in the field of Electronics and Communication Engineering by infusing scientific temper in the students leading towards research and to grow as centre of excellence in the field. The vision of the department is to provide education to students that is directly applicable to problems and situations encountered in real life and thus foster a successful career. The department aims to provide the best platform to students and staff for their growth.



From HOD's Desk:

Welcome to the Department of Electronics and Communication Engineering, VNIT Nagpur. The department started with an intake of 30 in 1984.

Subsequently, the intake increased progressively, and the current intake for the UG program of B.Tech ECE is 125, and that of the PG program, namely M.Tech in Communication System Engineering, is 25. The department also offers PhD and Post-Doctoral programs. The department remains the most sought after within the first two choices. The department has a good number of highly qualified and experienced faculties. The average experience of the faculty is 15 years, and the Professors have experience of 25 to 35 years. The department has produced GATE/CAT toppers for various years and Institute Visvesvaraya Medalists for many years. The alumni are highly placed, with 5% being entrepreneurs. The IPR of the department is very good in terms of faculty research publications with high citations in Google Scholar and SCI-indexed Journals and granted patents. There are always a good number of externally funded or sponsored research projects ongoing. The department has a very active and fruitful institute-industry interaction.

Dr. Ashwin Kothari

Professor and Head of the Department

The department has the following centers of excellence: 1) Center of Excellence in Embedded Systems established with grants from World Bank TEQIP funds. The center continues to train aspiring students with evening training programs. 2) CoE of AI 3) 5G LAB Center 4) CoE of EMC-EMI. The department works in upcoming and niche areas of electronics and communication engineering like 5G/6G, RF and Antennas, Signal Processing, AI/ML, Optoelectronics and Photonics, IoT, Automotive Electronics, Sensors and Instrumentation, Microelectronics, Biomedical Signal and Image Processing, Image Processing and Pattern Recognition.

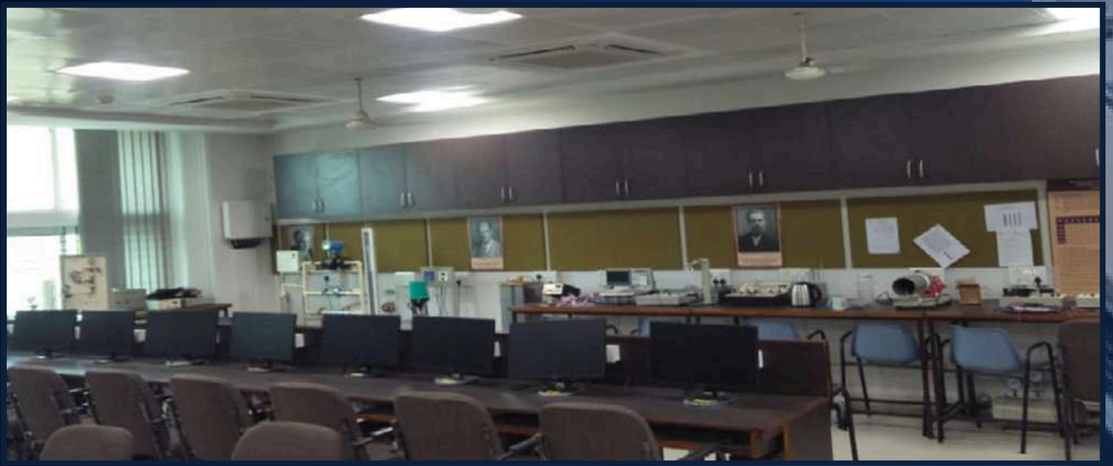
The department lays emphasis on clear fundamental concepts and applications of engineering through mini and major projects and internships. It is based on NEP 2020 and is accommodating for slow as well as fast learners. The annual external academic audit is a quality control exercise to keep abreast with technology and academic standards. Both the UG and PG programs are accredited for six years. The labs in the department have the latest equipment and students are made fully conversant with it with a considerable hands-on component. On the consultancy front, the department is growing now. The synergy of the department is very high. The department also offers an online PG program of M.Tech in Applied AI for working professionals. We are very confident that our students will excel in their performance wherever they go.

- Dr. Ashwin Kothari

STUDENT ACTIVITIES

Department of Electronics and communication provides an environment that nurtures leaders in addition to sound engineers. Notwithstanding the tight academic schedule that students have, they manage to indulge in various extra-curricular activities. Students from our department manage host of activities ranging from interdepartment competitions and national level festivals to managing facilities like messing and hostel infrastructure. These activities provide multitude of holistic improvement platforms for students to hone their technical and managerial skills, which are imperative for one to excel in his/her workplace.

OUR LABS



STUDENT ACHIEVEMENTS

- Gunjan J Paliwal published papers at the ISAP conference in Kuala Lumpur, Malaysia, titled "SIW Cavity-Backed Slot Antenna Using High-Order Radiation Modes for Broadband Applications in X Band" and at the WAMS conference in Visakhapatnam. She received the Best Poster Paper Award for "Bandwidth Enhancement of SIW Cavity Backed Slot Antenna Using High-Order Radiation Modes," respectively, on March 2, 2024.

- Dhanshree Warokar's paper, "Right Ventricular Ejection Fraction Prediction Using 2D Echocardiograms," was accepted in collaboration with IIT BHU.

Best Paper Award

Swadeshi microprocessor challenge

STUDENT ACHIEVEMENTS

Raj Kumar Thenua was a gold medalist in the "Introduction to Internet of Things (IoT)" course at IIT Kharagpur and delivered expert talks at IIT Madras on problem-based learning using MATLAB programming and at ADRDE, a pioneering R&D lab of DRDO, on 5G wireless communication.

We are glad to inform that Shankar Aenagandula published a paper on microwave non-destructive testing and evaluations at Taylor and Francis. Additionally, we are delighted to share that Suman Talla published two of his works in the IEEE Sensors Journal.

Drishti Diwani published a paper titled "QoS Aware Task Offloading for Digital-Twin Enabled Connected Vehicular Network," presented at the IEEE WCNC 2024 in Dubai, under Dr. Vivek Ashok Bohara and PhD scholar Ms. Akshita Gupta from IIIT Delhi.

Furthermore, Ashay Anand Atkar published the paper titled "Integrated Polarization Tracking and Control for Enhanced Performance in Polarization Multiplexed Self-Coherent Detection".

Another notable achievement comes from Prabhune Atharva Gajanan, who achieved a commendable GATE rank of 937 in the Electronics and Communication Engineering paper.

Faculty Acheivements

Electro Magnetic Interference and Electro Magnetic Compilance with a budget of 2.5 crore is in progress under the guidance of Dr.Ashwin Kothari, Dr. Abhay Gandhi, Dr.Joydeep Sengupta, Dr.Neeraj Rao, Dr.Arvind Kumar.

Under the supervision of Dr.Prabhat Sharma and Dr.Anamika Singh Bharat 5G Laboratory having budget of 1 crore is under progress.

Center for Artificial Intelligence (1.2 crore) has 56 computing systems with GPUs and 1 High -end system with 128GB RAM and 128GB GPU.Associated faculties are Dr. Kishor Bhurchandi and Dr.Deep Gupta

CoE in Combedded Systems(9crore) is coordinated by Dr.Ashwin Kothari .

Faculty Acheivements

Many faculties have received sponsered projects from agencies such as ISRO ,SERB ,SPARC grant , MEITY .

Many faculties such as Dr Prabhat Sharma, Dr arvind kumar have received international fellowships

Some faculties are rated as top 2% researcher.

Dst sire fellowship to Dr.Arvind Kumar.

For more details please visit departmental website:
https://vnit.ac.in/engineering/ece/

Electronics and Communication Engineering B.Tech

Technical Skills

C, C++, Java, Python, R, Kotlin , VHDL , Verilog, MATLAB

Machine Learning and Data Science:
Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch)
Deep Learning, Neural Networks, AI Tools

DATABASES :
SQL, MySQL, PostgreSQL, MongoDB, NoSQL, Firebase

WEB DEVELOPMENT:
HTML, CSS (Tailwind, Bootstrap),
JavaScript (React, Node.js, Express)

Web & Mobile Development: Flutter, React Native , Django, Android studio, RESTAPIS
Version Control and Containerization: Git, Github , Docker

Security: Cryptography, Steganography, Web Exploitation, Digital Forensics

Office and Productivity: POWER BI, **Microsoft Office(Advance excel), LaTeX, Canva, Adobe** (Photoshop, Illustrator), Google Analytics, **Tableau**

Electronics and Communication Engineering B.Tech

ENGINEERING DESIGN:

Embedded Systems: Arduino IDE, Embedded C, VHDL, Verilog, FPGA, RTL Design and Verification,Digi XBee Modules.

Analog and Digital Circuit Design: CMOS Design, PCB Design, Antenna Design and Analysis, CST Studio Suite

Signal Processing and Communication: Digital Communication, Wireless Communication, Audio/Image/Video Processing, MATLAB/Simulink, NetSim tool .

CAD/CAM: AutoCAD, Fusion 360, SolidWorks

Robotics and Automation: Arduino IDE,ROS, Gazebo, PyBullet, Issac Gym, IOT Embedded Development , Smart systems

Control Systems: Model Based System Design, Optimization

PCB Design Software: Eagle, Altium Designer, KiCad, EASYEda

VLSI Design Tools: Quartus, ModelSim , Xilinx, Cadence, Synopsys, Mentor Graphics , ngspice

Electronics and Communication Engineering B.Tech

COURSES

- Linear Network Theory
- Electronic Devices
- Signals and Systems Analysis
- Linear Algebra & Application
- Numerical Methods and Probability Theory
- Digital Signal Processing
- Analog Circuit Design
- Analog Communication
- Embedded Systems
- Measurement & Instrumentation
- Wave Guides and Antennas
- Device Modeling
- Digital Communication
- Digital Hardware Design
- RF & Microwave Engineering
- Machine Learning with Python
- Wireless Communication
- Wireless Sensor Network

- Optimal Systems
- Power Electronic Devices and Circuits
- Communication Networks
- Fundamentals of Information Theory
- Satellite Communication
- Advanced Microprocessors and Interfacing
- Biomedical Signal Processing
- Adaptive Signal Processing
- Radar Engineering
- Advanced Sensors and Instrumentation
- Computer Architecture & Organization
- CMOS Design
- Automotive Electronics
- Concepts in Operating Systems
- Digital Image Processing
- Advanced Digital Signal Processing
- Data Structures and Algorithms

Electronics and Communication Engineering M.Tech

Programming Languages & Coding platform :

Antenna and circuit designing software:
Cadence AWR, CST Microwave Studio, Proteus, Lab View

Network Analysis Tools : Wireshark, Cisco Packet Tracer , NetSim Tools

Analog & Digital Communication , Digital circuit kit , FPGA , Embedded System kits

Advanced Technological Competence:
Internet of Things, Artificial Intelligence, Machine Learning, Deep Learning

Model-Based Design and Simulation Software : MATLAB Simulink

Electronics and Communication Engineering M.Tech

COURSES

Signal Processing for Communication System
Statistical Signal Analysis
Information Theory and Coding
Wireless Channels
Computational Electromagnetism
Fuzzy Logic and Neural Networks
Communication Theory
Converged Communication Networks
Radio Frequency Circuit Design
Electromagnetic Interference and Compatibility
Topics In Communication Systems
Photonic Integrated Circuit and Communication System

Electronics and Communication Engineering PHD

AREAS OF SPECIALSATION & WORK OF INTEREST

Our ECE department proudly presents PhD candidates excelling in:

- Medical Image Processing
- No-Reference Video Quality Assessment
- NOMA, CR, AmBC, RIS, Optimization
- Molecular & Wireless Communications

Specialized Expertise

- RF & Microwave Engineering: Frequency Selective Surfaces, EMI/EMC, SI/PI, Microwave NDE, Antennas, RF Design.
- Ambient Backscattering & RIS
- Video Processing & Computer Vision: Denoising, Deep Learning, NLP.
- EEG Signal Processing: Fatigue classification.
- Speech Processing
- Machine Learning & AI: Matlab for Signal & Image Processing.
- IoT & WSNs
- Deep Learning for CAD

PROMINENT RECRUITERS

IT

magicbricks
D E Shaw & Co
LOWE'S
VISA
APPDYNAMICS
Goldman Sachs
Gameskraft
wipro
TRILOGY INNOVATIONS
JPMORGAN CHASE & CO.
Morgan Stanley
CREDIT SUISSE
HILTI
amazon
Fidelity INVESTMENTS
incedo
AT Motion Reimagined.
WELLS FARGO
airtel
sling
DECIMAL POINT Innovative Research Solutions
ORACLE
citi
Cognizant
UBS
Netcracker
accenture
un dos tres
Jio
celigo
SAMSUNG
PayU
Schneider Electric
NCR
Capgemini
OLA ELECTRIC
OPTUM
ICICI Bank
rapido
HDFC BANK

PROMINENT RECRUITERS

CORE

Ford
adani
intel.
Hitachi Energy
NXP
MARVELL
CATERPILLAR
ExxonMobil
vedanta
Texas Instruments
BAJAJ
THERMAX
DASSAULT SYSTEMES
SIEMENS
AU SMALL FINANCE BANK
AM/NS INDIA
Reliance Industries Limited
BOSCH
Deloitte.
PHILIPS
SUN PHARMA
TATA
EXL
IQVIA
EATON
iQUANTI
AXXELA
EY
KPMG
NOVARTIS
09
ZS
fractal
Hindustan Unilever Limited
INCREFF
merilytics

PSU's

BHARAT ELECTRONICS
इंडियनऑयल IndianOil
Bharat Petroleum
गेल GAIL
HPCL GAS
सी-डॉट C-DOT
ASHOK LEYLAND

MANAGEMENT

Faculty Corner

Professors

Dr. Ashwin Kothari

Areas Of Specialization:
Communication, Signal processing, Rough Sets, Cognitive Radio, Reconfigurable Antennas Commbeded Systems: Hybridization Of Communication and Embedded Systems.

Dr. A. S. Gandhi

Areas Of Specialization: Wireless communication, RF circuits and systems, Computer networks

Dr. K. M. Bhurchandi

Areas Of Specialization:
Embedded Systems, Image Processing

Faculty Corner

Associate Professor

Dr. Vishal R. Satpute

Areas Of Specialization:

Image Processing, Conputer Vision, Signal Processing, Cryptography, Bio Metrics

Faculty Corner

Assistant Professors

Dr. Saugata Sinha

Areas Of Specialization:

Image Processing, Pattern Recognition, Medical Imaging. Signal Processing.

Dr. Pradnya H. Ghare

Areas Of Specialization:

Body Area Networks, Wireless Sensors Network,
Analog Circuit Design,EEG Signal Processing

Dr. Prabhat Kumar Sharma

Areas of Specialization

Physical Layer aspects of Wireless Communications, Molecular and Biological Communications

Dr. Deep Gupta

Areas Of Specialization:

Signal & Image Processing, Medical Image Analysis, Machine Learning, Computer Vision, AI in Healthcare, Food Image Analysis

Dr. Joydeep Sengupta

Areas Of Specialization:

Communication & Microwave.

Faculty Corner

Assistant Professors

Dr. Neeraj Rao

Areas Of Specialization:
Antennas and Microwave

Dr. K. Surender

Areas Of Specialization:
Control and Instrumentation

Dr. Anamika Singh

Areas Of Specialization:
Silicon Photonics Integrated Circuits, High Frequency Antennas, Optical Biosensors, Optical Communications

Dr. Ankit A. Bhurane

Areas Of Specialization:
Signal Image and Video Processing, Communications, Embedded Systems

Dr. Punitkumar Bhavsar

Areas Of Specialization:
Human-in-the-loop cyber physical systems

Dr.Praveen Pawar

Areas Of Specialization :
Wireless Communications

Faculty Corner

Assistant Professors

Dr. Amit Agarwal

Areas Of Specialization:

Wireless Communication, Optical Wireless Communication, Performance analysis of amplitude constrained channel, Energy Efficient transceivers design, Quantum Communication, Next generation of wireless systems.
ialization

Dr. Arvind Kumar

Areas Of Specialization:

Substrate Integrated Waveguide (SIW) Based Antennas, Millimeter wave and THz Technologies

Dr.Vipin Kamble

Areas Of Specialization:

Signal, Image and Video Processing

Contact Us

Placement Cell

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Associate Dean, Training & Placement

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Dr.Pradnya Ghare

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